

COMMUNICATION

[View Article Online](#)
[View Journal](#) | [View Issue](#)Cite this: *J. Mater. Chem. A*, 2019, 7, 15062Received 3rd May 2019
Accepted 5th June 2019

DOI: 10.1039/c9ta04609g

rsc.li/materials-a

Selectively tuning gas transport through ionic liquid filled graphene oxide nanoslits using an electric field†

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Hybrid membranes with nanoconfined ionic liquid (IL) have demonstrated promising potential for CO₂ separation, while the trade-off between their permeance and selectivity is still an obstacle. In this work, an IL is confined in the 2D nanochannels (nanoslits) between graphene oxide (GO) nanosheets to form a GO-supported IL membrane (GO-SILM). By applying an external electric field (EEF), the separation performance is significantly elevated, doubling the CO₂ permeance to 260 GPU and enhancing the selectivity of CO₂/H₂, CO₂/CH₄, and CO₂/N₂ up to 43, 262, and 529, respectively, remarkably higher than that of a GO-SILM without an EEF. The performance is superior to the upper bound of state-of-the-art membranes. Molecular dynamics simulations show that the breakthrough comes from EEF-induced redistribution of cations and anions of the IL in the GO nanoslits, and thus changes the adsorption/desorption free energy, free volume and interaction energy of the IL. It has tapped the potential of the IL to its best. The synergistic effect of solubility and diffusivity of CO₂ results in asymmetric CO₂ permeance under positive and negative EEFs. These findings may open up an avenue for the design of advanced membranes for gas separation.

Membrane-based gas separation has great potential to meet the insistent demand of capturing and enriching CO₂, one of the greenhouse gases.¹ However, the intrinsic permeance–selectivity trade-off requires tremendous efforts to make a breakthrough.^{2,3} One of the common solutions is to expand the selection of membrane materials. Highly-ordered porous metal–organic frameworks (MOFs),^{4–6} covalent organic

frameworks (COFs),⁷ atomic-thick two-dimensional (2D) materials,^{8,9} ionic liquid (IL)-based membranes,^{10,11} and functionalized polymers, as well as their composites^{12–15} have been explored to construct cost-effective and energy-efficient membranes. However, there is a performance ceiling obtained by these approaches, which is due to the intrinsic material properties.

Another approach is to utilize the virtue of external stimuli, such as temperature, external pressure and external electric field (EEF), which can tune the pore size or interlayer spacing of the membrane, or change the interaction between the membrane and target guests.^{16–22} However, temperature- or pressure-control is usually followed by significant energy consumption, as they always need much higher temperature or pressure than usual, as well as low accuracy and a narrow range of tuning. Nonetheless, none of these abovementioned drawbacks become a problem when using an EEF as the stimulus. Previous research studies indicate that the permeability and selectivity of ions in aqueous solution through graphene nanopores can be modulated by the voltage because of a voltage-dependent transport barrier.^{23,24} Recent experiments show that a low voltage can modulate ionic diffusion through graphene-based membranes by modifying the interfacial electrical double layer (EDL).²⁵ Moreover, an EEF can switch the gas permeance through zeolitic imidazolate framework-8 (ZIF-8) by modifying the pore stiffness through structural transformation.²⁶ Except for these host membranes, the EEF can also tune the performance through the target guests, such as EEF-enhanced dissociation of interlayer water in the graphene oxide (GO) channels.²⁷ EEF is thus demonstrated to be a convenient and promising way to improve the separation performance without involving complex chemical modifications.

ILs have been proposed as potential absorbents because of their excellent properties including low volatility, high thermal stability and physicochemical tunability.²⁸ Supported IL phases,²⁹ porous substrate supported IL membranes³⁰ and 2D nanochannel confined IL membranes^{31–33} have been developed

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† Electronic supplementary information (ESI) available. See DOI: 10.1039/c9ta04609g

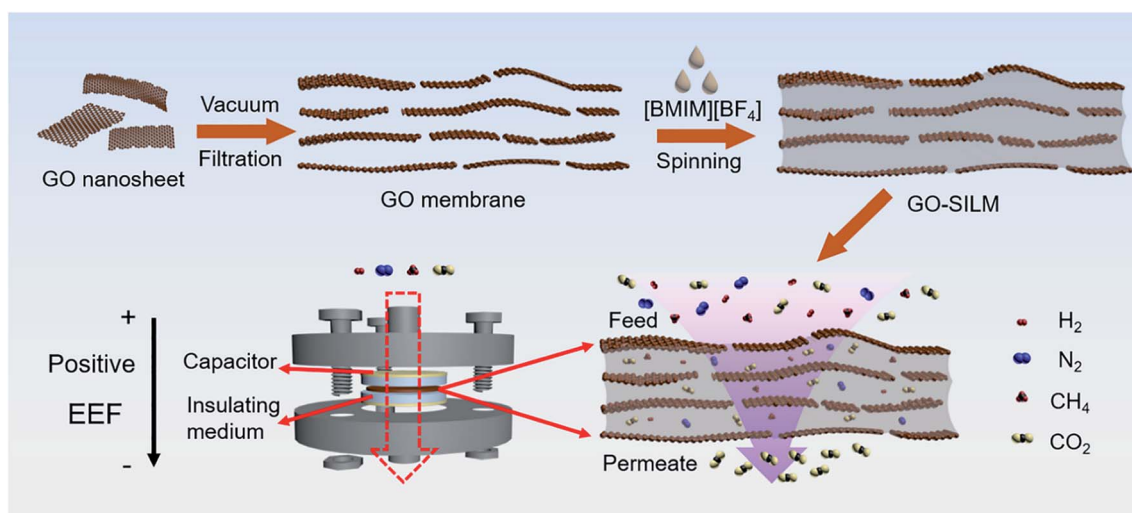
‡ W. Y., K. Z. and Q. H. contributed equally to this work.

to separate and capture CO_2 . However, there still may be room for improvement of the performance, and new strategies should be introduced. In contrast to water and organic liquids consisting of charge-neutral molecules, ILs are a neutral assembly of charged ions, so an EEF will drive the movement of anions and cations,³⁴ and in turn modify the spatial distribution. Moreover, the motion of ions will change the vacancies between them and further modulate the motion of gas molecules. Previous results indicate that gases have different affinity to anions or cations.^{31,35} Consequently, the gas separation performance of IL-based membranes will be changed under an EEF. Experiments have also confirmed that an EEF will drive the formation of an EDL^{36,37} and the generation of the electrorheological effect.³⁸ In brief, we believe that use of EEF is a potential method to improve the gas separation performance.

In this work, we applied an EEF on IL filled nanoslits, namely GO supported IL membranes (GO-SILMs), to study the changes in the properties of nanoconfined IL and the gas separation performance of GO-SILM. Our molecular dynamics (MD) simulation-based analysis shows that the redistribution of anions and cations of the IL in GO nanoslits under an EEF leads to opposite changes in the CO_2 solubility under a positive or negative EEF. However, both positive and negative EEF enhance the diffusivity of CO_2 . The above two factors result in asymmetric CO_2 permeance through the nanoslits in GO-SILMs under positive and negative EEFs. In contrast, the effect of the EEF on permeance is minor or even converse for H_2 , CH_4 and N_2 . Both the permeance and selectivity of CO_2 molecules of GO-SILMs are significantly improved under the EEF, demonstrating doubled permeance and 5 times higher CO_2/N_2 selectivity. The performance is further shown to be recyclable by applying and removing the EEF. This is promising to tune gas permeation and design smart gas separation membranes.

A GO dispersion was prepared by a modified Hummers method, as mentioned elsewhere,³¹ and the size of the resulting GO nanosheet is about $1\ \mu\text{m}$, with a height of $0.7\ \text{nm}$ (Fig. S1†).

The GO membrane was prepared by vacuum filtration through a porous anodic alumina oxide (AAO) membrane or polycarbonate membrane, with a pore size of $200\ \text{nm}$. The X-ray diffraction (XRD) pattern in Fig. S2† shows that the d -spacing of the GO membrane is about $0.89\ \text{nm}$. Based on the good wettability between GO sheets and the IL, 1-butyl-3-methyl imidazolium tetrafluoroborate $[\text{BMIM}][\text{BF}_4]$,³¹ the GO-SILM was prepared by dropping a certain volume of $[\text{BMIM}][\text{BF}_4]$ on the GO membrane for 2 h, and then spinning (Scheme 1). The wrinkles on the surface of the resulting GO-SILM are shorter (Fig. 1a and b). Fig. 1d shows that the laminated structure of the original GO membrane (Fig. 1c) disappears due to the incorporation of $[\text{BMIM}][\text{BF}_4]$. By controlling the filtered volume of the GO dispersion, GO-SILMs with different thicknesses were prepared as shown in Fig. S3.† The thinnest GO-SILM membrane reaches $72\ \text{nm}$, which contains $0.01\ \text{mg}$ GO, as shown in Table S1.† The mass ratio of GO and $[\text{BMIM}][\text{BF}_4]$ is 0.43 ± 0.03 , with an increment of $120\% \pm 10\%$ in thickness after infusing $[\text{BMIM}][\text{BF}_4]$, and the interlayer distance shifts from 0.89 to $2.11\ \text{nm}$ (the calculation can be seen in our previous work³¹). Fig. S4† shows the Fourier transform infrared (FTIR) spectra of GO-SILMs and peak shifts are listed in Table S2,† which indicate the confinement of $[\text{BMIM}][\text{BF}_4]$.³¹ The peaks of both anions and cations have shifts, which are related to the interaction with GO nanosheets, and the peak shifts of the imidazole ring are more distinct, suggesting a stronger interaction. Differential scanning calorimetry (DSC) curves in Fig. S5† show that the freezing point increased around $10\ ^\circ\text{C}$, which also demonstrates $[\text{BMIM}][\text{BF}_4]$ being nanoconfined in GO nanoslits. The uniform distribution of $[\text{BMIM}][\text{BF}_4]$ in GO nanochannels is further confirmed by energy dispersive X-ray spectroscopy (EDS) (Fig. 1e–h), because N, B and F elements distribute uniformly along the cross section of GO-SILMs. The EEF (-5 – $5\ \text{V mm}^{-1}$) is caused by applying a certain direct current (DC) voltage to capacitor plates, which are sandwiched between the sides of the membrane (not in contact), just as Scheme 1 shows, to tune the gas permeance.



Scheme 1 The process of preparation and testing of a home-made test device of a GO-SILM.

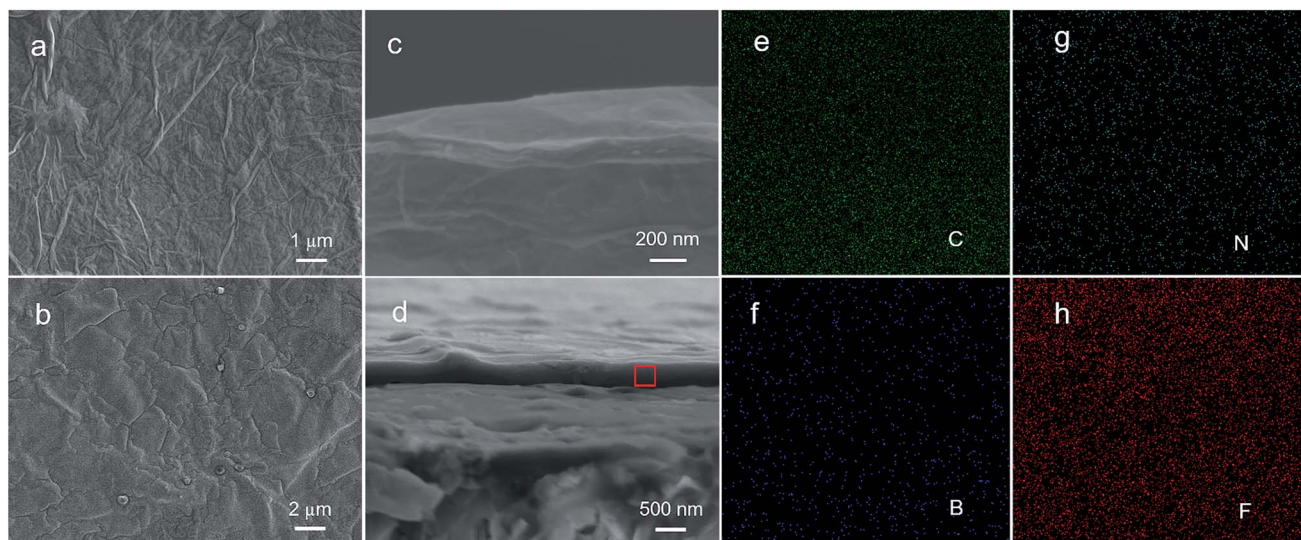


Fig. 1 The SEM images of the GO membrane and GO-SILM and the EDS spectrum of the GO-SILM. (a and b) Surface of GO and the GO-SILM; (c and d) cross section of GO and the GO-SILM; and (e–h) the EDS mapping images of C, B, N, and F recorded from the red pane in (d), respectively.

As mentioned above, ILs are liquid electrolytes made of ionic molecules. With an EEF applied, cations and anions are forced to move in opposite directions. For the GO-SILM studied in this work, part of the anions will be driven to move to the feed side of GO nanoslits under a positive EEF, while part of the cations move to the permeate side. In order to investigate the response of the IL under an EEF, we perform MD simulations by using an IL film model (see the ESI† for details). The results indicate that the separation between cations and anions increases with the strengthening of the EEF (Fig. 2), and the displacement of $[\text{BF}_4]^-$ is more significant than that of $[\text{BMIM}]^+$ due to its smaller molecular size. This conclusion applies to negative EEFs with opposite displacements. As for the spatial distribution of the IL in the GO-SILM, the results will be far more complex than our simple model because of the complex microstructural features and chemistry of GO nanoslits, while the conclusion that the distribution of ILs in the GO-SILM can be adjusted by an EEF is expected to hold.

In the solution-diffusion process of gas separation, the permeability is calculated as $P = D \times S$, where D and S are the diffusivity and solubility of gas molecules in the IL.³⁹ The permeance can be enhanced by elevating D or S and the influence of nanoconfinement of the IL on D and S .³¹ The anion affinity of CO_2 in the IL has been demonstrated, in contrast to N_2 , H_2 and CH_4 .^{31,35} Applying a positive EEF, the absorption of CO_2 increases as more $[\text{BF}_4]^-$ molecular ions accumulate on the feed side of the GO-SILM. However, on the permeate side, less $[\text{BF}_4]^-$ ions exist, which results in faster desorption of CO_2 . In contrast, the absorption and desorption of CO_2 will be decreased under a negative EEF. Fig. 3a shows the changes in the adsorption/desorption free energy, G , (see the ESI† for details) of CO_2 under an EEF using an IL film model (Fig. 2a), which increases while declines under negative EEFs. The changes in G for H_2 , CH_4 and N_2 show reverse dependence on the EEF as they have more affinity to cations rather than

anions.³¹ It was reported that the gas solubility is positively correlated with the free energy of sorption,⁴⁰ and thus the MD simulation results suggest that the EEF can facilitate CO_2 permeance through the GO-SILM by boosting S . Moreover, the redistribution of the IL (Fig. 2b) also results in the expansion of free volume (see the ESI† for details) in the 2D GO nanoslits (Fig. 3b), which is another important factor that determines the solubility. The free volume increases with the strengthening of the electric field under both positive and negative EEF. Research shows that the increase of free volume leads to a higher gas solubility.⁴¹ So the gas permeance will be further enhanced. Fig. 3c clearly shows the negative correlation between anion–cation interactions and the electric field under both positive and negative EEF. As known, the weaker cation–anion pair attraction allows a lower viscosity⁴² and thus promotes gas diffusion.⁴³ In summary, at the gas–IL interface, the adsorption/desorption process has a reverse influence on S because of the EEF, which is enhanced under a positive EEF while weakened under a negative EEF. On the other hand, inside the nanoslits, the change of free volume of the IL and anion–cation interactions will enhance both S and D , no matter under a positive or negative EEF. Based on our analysis, the EEF can enhance the permeability of CO_2 . As for the selectivity of CO_2 , the adsorption/desorption process can facilitate it while influences from the change of free volume and/or anion–cation interactions are unknown because of the lack of gas-specific information.

To assess the predictions from simulations, we investigate the gas separation performance of the prepared GO-SILM under an EEF using a home-made separation system as shown in Scheme 1, under an applied pressure of 0.16 MPa, with an effective area of 2.2 cm^2 at 273 K. Tested gases are single pure gas. A positive EEF is set while the anode is applied on the feed side, and a negative EEF is just in reverse. Fig. S6† shows that with the increase of membrane thickness, the gas permeance

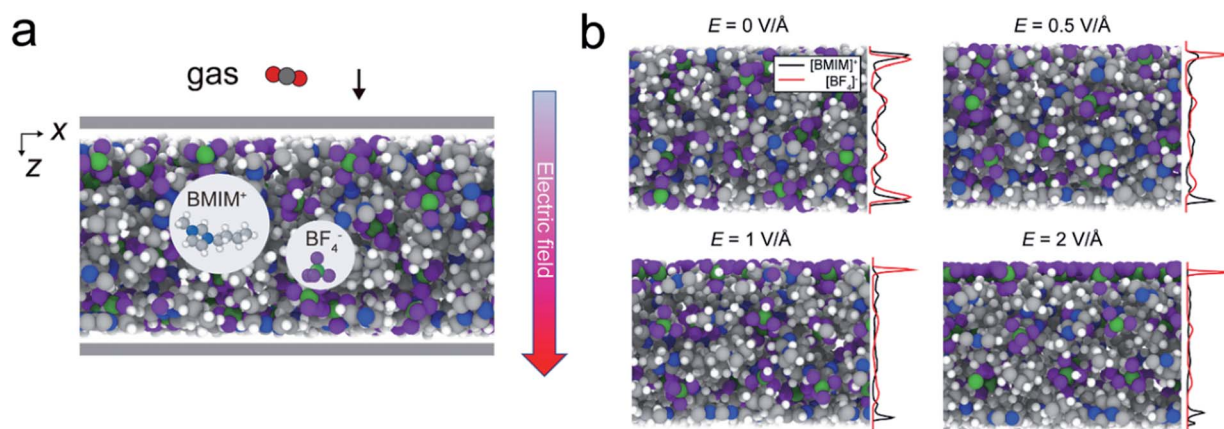


Fig. 2 (a) The model of the IL film under an EEF; (b) the distribution of $[BMIM]^+$ and $[BF_4]^-$ in the 2D channel confined between GO sheets under an EEF.

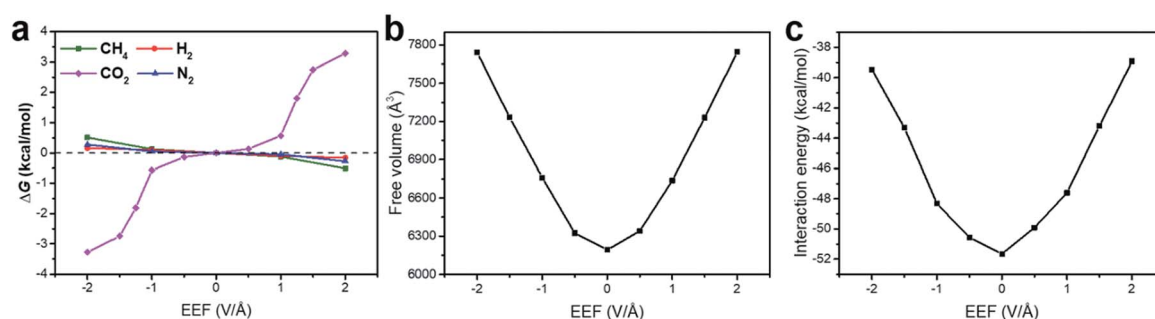


Fig. 3 (a) The change of adsorption/desorption energy of gases (G) in $[BMIM][BF_4]$ confined in GO-SILMs under an EEF; (b) the free volume of the IL under an EEF; and (c) the anion-cation interaction under an EEF.

through GO-SILMs decreases, while the selectivity increases, in agreement with the tendency as reported elsewhere.³¹

In this work, we choose the GO-SILM with a thickness of 380 nm (Fig. 1) which contains 0.05 mg GO to investigate the performance under an EEF. The capacitor plates were not directly put on the GO-SILM to eliminate the Joule effect,⁴⁴ as there was no electric current flowing through the GO-SILM. The results are shown in Fig. 4. Fig. 4a indicates that permeances of H₂, CH₄ and N₂ through the GO-SILM have a very slight downward trend with the increment of the positive EEF. But the negative EEF has almost no effect on the permeances of N₂ and H₂, while they show a slightly downward trend for CH₄. For CO₂, the permeance through the GO-SILM obviously increases no matter under a positive or negative EEF. But the magnitude of increment under a positive EEF is larger than that under a negative EEF, which shows asymmetry in permeance, and the maximum permeance is doubled to 260 GPU compared with the GO-SILM without an EEF. For comparison, the permeance of these gases through the GO membrane under an EEF was investigated and the results are shown in Fig. 4b. The permeance of H₂ has some increase under both positive EEF and negative EEF, while the permeance of CO₂ only increases under a positive EEF, and the others have little changes. As a result, the selectivity has the same trend with permeance of CO₂.

Anyway, the separation selectivity through the GO membrane under an EEF is still close to the Knudsen selectivity,⁴⁵ which is too low for separating CO₂ from other three light gases. Above all, the gas permeance asymmetry, especially CO₂ permeance is matched with our stimulation result. Accordingly, the selectivity of CO₂/H₂, CO₂/CH₄ and CO₂/N₂ shows the same trend with CO₂, that is, it increases both under a positive and

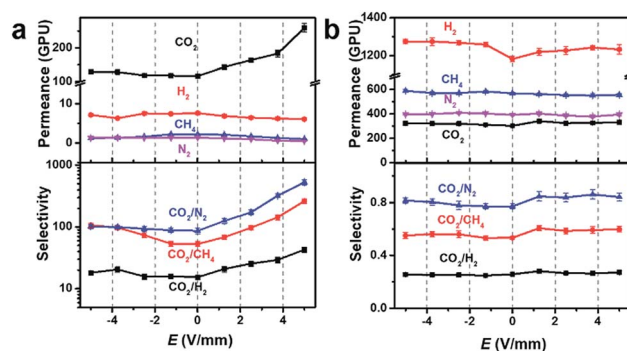


Fig. 4 Gas separation performance of the GO membrane and GO-SILM both including 0.05 mg GO. The permeance and selectivity of (a) GO-SILM and (b) GO membrane, respectively, under an EEF; error is within $\pm 15\%$.

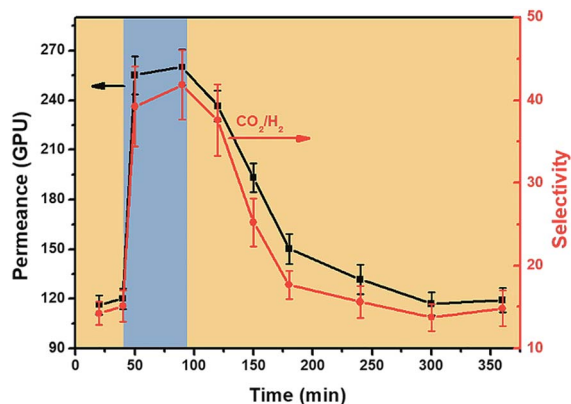


Fig. 5 The permeance and selectivity of CO_2/H_2 in the process of applying (blue zone) and removing (yellow zone) EEF.

negative EEF, and the magnitude is larger under a positive EEF. The CO_2 permeance of the GO-SILM under a positive EEF is 260 GPU and the selectivity of CO_2/H_2 , CO_2/CH_4 and CO_2/N_2 is enhanced to 43, 262, and 529, which are 2, 4 and 5 times larger than that of the GO-SILM without an EEF, respectively. This performance is better than most other membranes as shown in Fig. S7 and Table S3.† The performance of the GO-SILM is independent of the porous substrate as GO-SILMs prepared on a porous polycarbonate (PC) membrane also have similar performance (Fig. S8†). The improvement of performance under an EEF is also applicable to other IL filled GO nanoslits, such as 1-ethyl-3-methylimidazolium tetrafluoroborate [EMIM][BF_4] (Fig. S8†).

Furthermore, the reversibility of the GO-SILM upon applying and removing the EEF was investigated (Fig. 5). The permeance of CO_2 rises to the maximum in ten minutes when applying an EEF and returns to its original permeance after removing the EEF for about 2–3 h with exponential decay. The response of removal is reminiscent of the relaxation process of a capacitor. The corresponding XPS results of GO membranes (Table S4†) before and after applying the EEF indicate that the GO membrane has not been reduced by EEF.

Conclusions

By applying an EEF, we greatly improve the performance of GO-SILMs for CO_2 separation compared to those without an EEF. The permeance of CO_2 improved from 117 GPU to 260 GPU under a positive EEF, and the selectivity of CO_2/H_2 , CO_2/CH_4 and CO_2/N_2 enhanced to 43, 262, and 529, which are 2, 4 and 5 times larger than those of GO-SILMs without an EEF, respectively. The redistribution of the IL in GO nanoslits can be tuned by an EEF and results in different adsorption/desorption free energies, free volumes and interaction energies, which are related to gas solubility and diffusivity. The performance is reversible when applying and removing the EEF. In a word, applying external stimuli, such as an EEF, is a convenient method to break the trade-off between permeance and selectivity of GO-SILMs without complicated chemical treatments.

Conflicts of interest

There are no conflicts to declare.

Acknowledgements

This work is supported by the National Natural Science Foundation of China (NSFC 21671171, 21875212, 51632008), the National Basic Research Program of China 973 Program (2015CB655302), the National Key Research and Development Program (2016YFA0200204), and the Zhejiang Natural Science Foundation (LD18E02001).

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